

YEAR 7 MATHEMATICS · CHAPTER 3

Whole numbers

3.01 Rounding and estimation**3.03** Division**3.05** Long division**3.07** Index notation and powers**3.09** Prime and composite numbers**3.11** Highest common factor**3.02** Multiplication**3.04** Divisibility tests**3.06** Roman numerals**3.08** Square roots and cube roots**3.10** Prime factors**3.12** Lowest common multiple

Name _____

Class _____

Teacher _____

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1 , then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

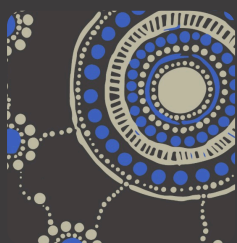
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In this chapter you will

- round and estimate
- multiply and divide whole numbers
- use divisibility tests
- use long division and Roman numerals
- use index notation, square roots and cube roots
- find prime numbers and prime factors
- find the HCF and LCM of two numbers

NSW Mathematics K–10 Syllabus (2022), Stage 4 Number and algebra: MA4-CAL-C-01 compares, orders and calculates with integers, and MA4-IND-C-01 (index notation, primes, factors and multiples).



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
3.01

Rounding and estimation

Date _____

Learning intentions

We are learning to:

- round whole numbers
- estimate answers by rounding first

Success criteria

I can:

- ☐ find the **rounding digit** and look at the digit after it
- ☐ round to the nearest 10, 100, 1000 or 10 000
- ☐ estimate by rounding each number first

Key terms Write the definition as your teacher explains it.

Round

Estimate

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Round 87 531 to the nearest thousand.

Example 2

Round 87 531 to the nearest ten.

YOU DO

On your own: Set A. Round each number to the nearest ten.

1 47

2 152

3 3875

4 995

5 6004

6 87 531

YOU DO On your own: Set B. Round to the place shown.

1 748 (hundred)

4 87 531 (ten thousand)

2 87 531 (hundred)

5 65 250 (thousand)

3 3499 (thousand)

6 149 999 (ten thousand)

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Estimate $425 + 81 + 940 + 22$.**Example 4** Estimate $864 + 250 - 417$.**YOU DO** On your own: Set C. Estimate by rounding each number to its first digit.1 $612 + 285$ 3 $38 + 71 + 405$ 5 $7210 + 2890$ 2 $4190 - 1820$ 4 $783 - 219 + 96$ 6 $5125 - 3980$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Estimate 93×19 .
Example 6 Estimate $2402 \div 32$.
YOU DO On your own: Set D. Estimate, then check with a calculator.**1** 48×21

3 $3950 \div 38$

2 612×9

4 A school has 28 classes of 31 students. Estimate the number of students.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Round 6382 to the nearest hundred.

Level 2Estimate $512 + 289 - 97$.
Level 3Estimate $4180 \div 21$.

Extension · Stretch yourself

Worked example

A number rounded to the nearest hundred is 4700. What are the smallest and largest whole numbers it could be?

- Numbers from 4650 round up to 4700
- 4750 rounds up to 4800, so stop at 4749

Answer: 4650 and 4749

YOU DO

On your own: read the worked example, then try these.

- E1** A number rounded to the nearest thousand is 12 000. Find the smallest and largest whole numbers it could be.

- E2** A crowd was “about 35 000” to the nearest thousand. What is the largest the crowd could have been?

Lesson summary

Missed the lesson? Start here.

Remember

- Find the rounding digit.
- Look at the digit to its right.
- 5 or more: round up. 4 or less: stay.
- Estimate: round first, then calculate.

Level 1

Round 87 531 to the nearest thousand

- Thousands digit 7, next digit 5
- Round up

Answer: 88 000

Level 2

Round 3499 to the nearest thousand

- Next digit 4

Answer: 3000

Level 3

Estimate 93×19

- $\approx 90 \times 20$

Answer: 1800

Extension

Nearest hundred is 4700

- 4650 to 4749

Answer: 4650 and 4749

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LESSON
3.02

Multiplication

Date _____

Learning intentions

We are learning to:

- multiply whole numbers mentally and on paper

Success criteria

I can:

- ☐ use mental strategies such as doubling or splitting a number
- ☐ multiply by a one-digit number on paper
- ☐ multiply by a two-digit number on paper

Key terms Write the definition as your teacher explains it.**Product****Algorithm**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** 12×4 **Example 2** 25×8 (use doubling)**YOU DO** On your own: Set A. Evaluate each product mentally.1 12×4 4 11×8 2 15×6 5 25×4 3 20×9 6 50×7

YOU DO On your own: Set B. Evaluate using a strategy.

1 34×5

4 16×5

2 25×8

5 45×4

3 33×9

6 21×11

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** 34×5 (use $\times 10 \div 2$)**Example 4** 33×9 (use $\times 10 - 1$)**YOU DO** On your own: Set C. Evaluate each product on paper.

1 126×7

3 375×6

5 1234×3

2 208×4

4 93×8

6 459×9



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** 126×7 **Example 6** 565×18 **YOU DO** On your own: Set D. Evaluate each product on paper.**1** 565×18 **3** 312×45 **2** 47×23 **4** A bus holds 57 people. How many people can 24 buses hold?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Evaluate 14×5 .
Level 2Evaluate 243×6 .
Level 3Evaluate 128×27 .

Extension · Stretch yourself

Worked example

Find the missing digits: $4\square \times 6 = 2\square 8$

- Units: $\square \times 6$ ends in 8, so $\square = 3$ or 8
- $43 \times 6 = 258$ ✓; $48 \times 6 = 288$ ✓

Answer: $43 \times 6 = 258$ or $48 \times 6 = 288$

YOU DO

On your own: read the worked example, then try these.

- E1 Find the missing digit: $\square 7 \times 5 = 185$.

- E2 Find two 2-digit numbers whose product is 1000.

Lesson summary

Missed the lesson? Start here.

Remember 1. Double and halve: $25 \times 8 = 50 \times 4$.
3. $\times 9$ is the same as $\times 10 -$ the number.

2. $\times 5$ is the same as $\times 10 \div 2$.
4. On paper: multiply each digit and carry.

Level 1

$$12 \times 4$$

- $10 \times 4 + 2 \times 4$

Answer: 48

Level 2

$$34 \times 5$$

- $340 \div 2$

Answer: 170

Level 3

$$565 \times 18$$

- $565 \times 8 = 4520$
- $565 \times 10 = 5650$
- Add

Answer: 10 170

Extension

$$4\square \times 6 = 2\square 8$$

- Try 3 and 8

Answer: 43 or 48

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LESSON
3.03

Division

Date _____

Learning intentions

We are learning to:

- divide whole numbers mentally and using short division

Success criteria

I can:

- ☐ divide by 10, 100 and 1000 mentally
- ☐ use short division with a one-digit divisor
- ☐ write a remainder when the division is not exact

Key terms Write the definition as your teacher explains it.**Quotient****Remainder**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

$2400 \div 10$

Example 2

$144 \div 9$

YOU DO

On your own: Set A. Evaluate each quotient mentally.

1 $2400 \div 10$

4 $3500 \div 100$

2 $48 \div 6$

5 $144 \div 12$

3 $72 \div 8$

6 $63 \div 9$

YOU DO On your own: Set B. Evaluate each quotient mentally.

1 $144 \div 9$

4 $96 \div 3$

2 $90 \div 5$

5 $81\,000 \div 1000$

3 $120 \div 4$

6 $84 \div 7$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $620 \div 5$

Example 4 $528 \div 4$

YOU DO On your own: Set C. Use short division.

1 $620 \div 5$

3 $1296 \div 8$

5 $2115 \div 9$

2 $528 \div 4$

4 $847 \div 7$

6 $936 \div 6$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $1296 \div 8$ **Example 6** $847 \div 7$ **YOU DO** On your own: Set D. Divide. Write any remainder.**1** $475 \div 6$ **3** $5000 \div 7$ **2** $1003 \div 4$ **4** 250 students go on buses that hold 45. How many buses are needed?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO**Show off your skill**Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Evaluate $360 \div 10$.
Level 2Evaluate $738 \div 6$.
Level 3Evaluate $1250 \div 8$, with remainder.

Extension · Stretch yourself

Worked example

I am a number between 50 and 70. Divided by 4 I leave remainder 1; divided by 5 I leave remainder 3. What am I?

1. Remainder 1 $\div$ 4: 53, 57, 61, 65, 69
2. Remainder 3 $\div$ 5: 53, 58, 63, 68

Answer: 53

YOU DO

On your own: read the worked example, then try these.

- E1** Find a number between 20 and 40 that leaves remainder 2 when divided by 3, 4 and 6.

- E2** $1000 \div 7$: what is the remainder?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. $\div$ 10, 100, 1000: move digits right.
 3. A remainder is what is left over.

2. Short division: divide each digit, carry the remainder.
4. In a problem, decide whether to round up.

Level 1

$$2400 \div 10$$

1. Move digits 1 place right

Answer: 240

Level 2

$$528 \div 4$$

1. $5 \div 4 = 1 \text{ r}1$
2. $12 \div 4 = 3$
3. $8 \div 4 = 2$

Answer: 132

Level 3

$$475 \div 6$$

1. $47 \div 6 = 7 \text{ r}5$
2. $55 \div 6 = 9 \text{ r}1$

Answer: 79 r 1

Extension

$$\div 4 \text{ r}1 \text{ and } \div 5 \text{ r}3$$

1. List and compare

Answer: 53

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LESSON
3.04

Divisibility tests

Date _____

Learning intentions

We are learning to:

- use divisibility tests for 2, 3, 4, 5, 6, 8, 9 and 10

Success criteria

I can:

- ☐ test for 2, 5 and 10 using the last digit
- ☐ test for 3 and 9 using the digit sum
- ☐ test for 4, 6 and 8

Key terms Write the definition as your teacher explains it.**Divisible****Digit sum**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Is 1208 divisible by 2? By 5? By 10?

Example 2

Is 1208 divisible by 3? By 9?

YOU DO

On your own: Set A. Write yes or no.

1 Is 346 divisible by 2?

2 Is 346 divisible by 5?

3 Is 780 divisible by 10?

4 Is 915 divisible by 5?

5 Is 4071 divisible by 2?

6 Is 5500 divisible by 10?

YOU DO On your own: Set B. Find the digit sum, then answer yes or no.

1 Is 471 divisible by 3?

4 Is 1000 divisible by 3?

2 Is 471 divisible by 9?

5 Is 2226 divisible by 3?

3 Is 5832 divisible by 9?

6 Is 8109 divisible by 9?

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Is 1208 divisible by 4? By 8?**Example 4** Is 1208 divisible by 6?**YOU DO** On your own: Set C. Test each number for 4, 6 and 8. Show your test.

1 324

3 516

5 2346

2 1208

4 7040

6 9112



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Which number is divisible by both 3 and 8: 30, 24, 90 or 94?
Example 6 Find the smallest digit $\square$ so that $52\square$ is divisible by 6.
YOU DO On your own: Set D. Use divisibility tests.**1** Test 2520 for 2, 3, 4, 5, 6, 8, 9 and 10.
3 Find all digits $\square$ that make $7\square4$ divisible by 3.
2 Which numbers are divisible by both 3 and 8: 30, 24, 90, 96, 120?
4 Can a number be divisible by 9 but not by 3? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Is 735 divisible by 5?

Level 2

Is 4815 divisible by 9?

Level 3

Is 1716 divisible by 6? Explain.

Extension · Stretch yourself

Worked example

Is 3 162 816 divisible by 12?

- $12 = 3 \times 4$, so test 3 and 4
- Digit sum $3 + 1 + 6 + 2 + 8 + 1 + 6 = 27$, divisible by 3
- Last two digits 16, divisible by 4

Answer: Yes

YOU DO

On your own: read the worked example, then try these.

- E1** Is 45 720 divisible by 15? Explain using tests.

- E2** Find the smallest 4-digit number divisible by 2, 3, 4, 5 and 6.

Lesson summary

Missed the lesson? Start here.

Divisibility tests

1. 2: even. 5: ends in 0 or 5. 10: ends in 0.	2. 3: digit sum divisible by 3. 9: digit sum by 9.
3. 4: last 2 digits by 4. 8: last 3 digits by 8.	4. 6: divisible by 2 and 3.

Level 1

$1208 \div 2?$

- Ends in 8, even

Answer: yes

Level 2

$1208 \div 3?$

- $1 + 2 + 0 + 8 = 11$

Answer: no

Level 3

$1208 \div 8?$

- $208 \div 8 = 26$

Answer: yes

Extension

$3\,162\,816 \div 12?$

- Test 3 and 4

Answer: yes

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LESSON
3.05

Long division

Date _____

Learning intentions

We are learning to:

- divide by a two-digit number using long division

Success criteria

I can:

- ☐ set out long division
- ☐ use the steps divide, multiply, subtract, bring down
- ☐ write the answer with a remainder

Key terms Write the definition as your teacher explains it.**Divisor****Dividend**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

$650 \div 25$

Example 2

$2976 \div 24$

YOU DO

On your own: Set A. Warm up: know your multiples.

1 12×7

2 25×6

3 24×4

4 15×8

5 32×3

6 16×9

YOU DO On your own: Set B. Evaluate.

1 $96 \div 12$

4 $240 \div 15$

2 $175 \div 25$

5 $128 \div 16$

3 $144 \div 24$

6 $99 \div 11$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $3079 \div 12$

Example 4 $864 \div 16$

YOU DO On your own: Set C. Use long division.

1 $650 \div 25$

3 $864 \div 16$

5 $3825 \div 15$

2 $2976 \div 24$

4 $1344 \div 12$

6 $2574 \div 11$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $5000 \div 32$ **Example 6** How many 45-seat buses are needed for 1000 people?
YOU DO On your own: Set D. Use long division. Write any remainder.**1** $3079 \div 12$ **3** $9876 \div 21$ **2** $5000 \div 32$ **4** 1000 people need 45-seat buses. How many buses?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO**Show off your skill**Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Evaluate $84 \div 14$.
Level 2Evaluate $1728 \div 16$.
Level 3Evaluate $4000 \div 27$ with remainder.

Extension · Stretch yourself

Worked example

A number divided by 23 gives 41 remainder 17.
Find the number.

1. Number = divisor $\times$ quotient + remainder
2. $23 \times 41 + 17 = 943 + 17$

Answer: 960

YOU DO

On your own: read the worked example,
then try these.

- E1** A number divided by 18 gives 55 remainder 4.
Find the number.

- E2** Find the largest 3-digit number that leaves
remainder 5 when divided by 13.

Lesson summary

Missed the lesson? Start here.

Remember

1. Divide: how many times does the divisor go in?
2. Multiply and write it under.
3. Subtract, then bring down the next digit.
4. Repeat. What is left is the remainder.

Level 1

$$96 \div 12$$

1. $12 \times 8 = 96$

Answer: 8

Level 2

$$650 \div 25$$

1. $65 \div 25 = 2 \text{ r}15$
2. $150 \div 25 = 6$

Answer: 26

Level 3

$$3079 \div 12$$

1. $30 \div 12 = 2 \text{ r}6$
2. $67 \div 12 = 5 \text{ r}7$
3. $79 \div 12 = 6 \text{ r}7$

Answer: 256 r 7

Extension

$$23 \times 41 + 17$$

1. $943 + 17$

Answer: 960

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.06

Roman numerals

Date _____

Learning intentions

We are learning to:

- read and write Roman numerals

Success criteria

I can:

- ☐ know the values of I, V, X, L, C, D and M
- ☐ add values when a symbol is followed by a smaller or equal one
- ☐ subtract when a smaller symbol comes before a larger one

Key terms Write the definition as your teacher explains it.**Numeral****Symbol**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

What is XXXVII?

Example 2

What is CCXL?

YOU DO

On your own: Set A. Write each Roman numeral as a number.

1 VIII

2 XII

3 XXXVII

4 LX

5 CCX

6 XV

YOU DO

On your own: Set B. Write each Roman numeral as a number.

1 IV

4 XL

2 IX

5 CCXL

3 XXIV

6 XC

Level 2 · Building up**WE DO**

With your teacher: copy what they write on the board.

Example 3

What is XXIV?

Example 4

Write 28 in Roman numerals.

YOU DO

On your own: Set C. Write each number in Roman numerals.

1 6

4 74

2 13

5 125

3 28

6 49



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Write 74 in Roman numerals.
Example 6 Write 125 in Roman numerals.
YOU DO On your own: Set D. Harder numerals.**1** Write 1999 in Roman numerals.
3 Write this year and your birth year in Roman numerals.
2 Write MMXXVI as a number.
4 Why is IC not used for 99?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write XVI as a number.

Level 2

Write 39 in Roman numerals.

Level 3

Write 2024 in Roman numerals.

Extension · Stretch yourself

Worked example

Evaluate $XLIV + LVI$ and write the answer in Roman numerals.

- $XLIV = 44$, $LVI = 56$
- $44 + 56 = 100$

Answer: C

YOU DO

On your own: read the worked example, then try these.

- E1** Evaluate $CCC - XLV$ and write the answer in Roman numerals.

- E2** Evaluate $XII \times IX$ in Roman numerals.

Lesson summary

Missed the lesson? Start here.

- Roman numerals**
- I 1, V 5, X 10, L 50, C 100, D 500, M 1000.
 - Larger then smaller: add ($VI = 6$).
 - Smaller before larger: subtract ($IV = 4$).
 - Never more than 3 of the same symbol in a row.

Level 1

XXXVII

- $10 + 10 + 10 + 5 + 1 + 1$

Answer: 37

Level 2

CCXL

- $200 + 40$

Answer: 240

Level 3

125

- $100 + 20 + 5$

Answer: CXXV

Extension

$XLIV + LVI$

- $44 + 56 = 100$

Answer: C

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.07

Index notation and powers

Date _____

Learning intentions

We are learning to:

- write repeated multiplication using index notation
- evaluate powers

Success criteria

I can:

- ☐ identify the **base** and **index**
- ☐ write a product like $3 \times 3 \times 3 \times 3$ as 3^4
- ☐ evaluate a power such as 5^3

Key terms Write the definition as your teacher explains it.

Base

Index

Power

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Write $3 \times 3 \times 3 \times 3$ in index notation.
Example 2 Write $7 \times 7 \times 7 \times 7 \times 7 \times 7 \times 7$ in index notation.
YOU DO On your own: Set A. Write in index notation.1 $4 \times 4 \times 4$

2 $6 \times 6 \times 6 \times 6 \times 6$

3 9×9

4 $2 \times 2 \times 2 \times 2 \times 2 \times 2$

5 $10 \times 10 \times 10 \times 10$

6 $5 \times 5 \times 5 \times 3 \times 3$

YOU DO On your own: Set B. Write as a product (expanded form).

1 8^3

4 2^4

2 3^5

5 7^1

3 11^2

6 $2^3 \times 5^2$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Evaluate 6^2 .**Example 4** Evaluate 5^3 .**YOU DO** On your own: Set C. Evaluate each power.

1 6^2

4 10^4

2 5^3

5 3^4

3 2^5

6 1^9



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Evaluate 10^5 .
Example 6 Evaluate 2^8 .
YOU DO On your own: Set D. Evaluate.**1** 2^8

3 $2^3 \times 3^2$

2 $4^3 - 3^3$

4 Which is larger: 2^5 or 5^2 ? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Write $7 \times 7 \times 7$ in index notation.
Level 2Evaluate 4^3 .
Level 3Evaluate $3^2 \times 2^4$.

Extension · Stretch yourself

Worked example

Write 64 as a power of 2, 4 and 8.

1. $64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$

2. $64 = 4 \times 4 \times 4$

3. $64 = 8 \times 8$

Answer: $2^6 = 4^3 = 8^2$

YOU DO

On your own: read the worked example, then try these.

E1 Write 81 as a power of 3 and of 9.

E2 Find n if $2^n = 1024$.

Lesson summary

Missed the lesson? Start here.

Remember 1. In 5^3 , 5 is the **base** and 3 the **index**.
3. Count the equal factors to find the index.

2. $5^3 = 5 \times 5 \times 5$, not 5×3 .
4. Squared means index 2; cubed means index 3.

Level 1

$3 \times 3 \times 3 \times 3$

1. Four 3s

Answer: 3^4

Level 2

5^3

1. $5 \times 5 \times 5$

Answer: 125

Level 3

2^8

1. 2, 4, 8, 16, 32, 64, 128, 256

Answer: 256

Extension

64 as powers

1. 2^6 , 4^3 , 8^2

Answer: 2^6 , 4^3 , 8^2

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.08

Square roots and cube roots

Date _____

Learning intentions

We are learning to:

- find square roots and cube roots

Success criteria

I can:

- ☐ know the square numbers to 15^2 and the cube numbers to 10^3
- ☐ find $\sqrt{\quad}$ as the number that squares to it
- ☐ find $\sqrt[3]{\quad}$ as the number that cubes to it

Key terms Write the definition as your teacher explains it.

Square root

Cube root

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

 $\sqrt{49}$

Example 2

 $\sqrt{144}$

YOU DO

On your own: Set A. Evaluate each square root.

1 $\sqrt{16}$

2 $\sqrt{49}$

3 $\sqrt{81}$

4 $\sqrt{100}$

5 $\sqrt{144}$

6 $\sqrt{1}$

YOU DO On your own: Set B. Evaluate each cube root.

1 $\sqrt[3]{8}$

4 $\sqrt[3]{125}$

2 $\sqrt[3]{27}$

5 $\sqrt[3]{1000}$

3 $\sqrt[3]{64}$

6 $\sqrt[3]{1}$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** $\sqrt[3]{8000}$ **Example 4** $\sqrt[3]{729}$ **YOU DO** On your own: Set C. Evaluate.

1 $\sqrt{400}$

4 $\sqrt{225}$

2 $\sqrt[3]{8000}$

5 $\sqrt{36} + \sqrt{64}$

3 $\sqrt[3]{729}$

6 $\sqrt[3]{216} \times \sqrt{4}$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $\sqrt{9 \times 16}$

Example 6 Between which two whole numbers is $\sqrt{50}$?
YOU DO On your own: Set D. Think it through.**1** Between which two whole numbers is $\sqrt{90}$?
3 Evaluate $\sqrt{(36 + 64)}$. Is it the same as $\sqrt{36} + \sqrt{64}$?
2 Between which two whole numbers is $\sqrt[3]{100}$?
4 A square has area 196 cm^2 . Find its side length.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Evaluate $\sqrt{64}$.
Level 2Evaluate $\sqrt[3]{125}$.
Level 3Evaluate $\sqrt{121} - \sqrt[3]{1000}$.

Extension · Stretch yourself

Worked example

Find $\sqrt{1600}$ without a calculator.

- $1600 = 16 \times 100$
- $\sqrt{16} = 4$ and $\sqrt{100} = 10$

Answer: 40

YOU DO

On your own: read the worked example, then try these.

- E1** Find $\sqrt{4900}$ and $\sqrt[3]{27\,000}$ without a calculator.

- E2** A cube has volume 343 cm^3 . Find the length of its edge.

Lesson summary

Missed the lesson? Start here.

Remember

- $\sqrt{49} = 7$ because $7^2 = 49$.
- $\sqrt[3]{64} = 4$ because $4^3 = 64$.
- Learn the squares to 15^2 and cubes to 10^3 .
- Estimate a root using the nearest squares.

Level 1

$$\sqrt{144}$$

- $12 \times 12 = 144$

Answer: 12

Level 2

$$\sqrt[3]{729}$$

- $9 \times 9 \times 9 = 729$

Answer: 9

Level 3

$$\sqrt{50}$$

- $49 < 50 < 64$

Answer: between 7 and 8

Extension

$$\sqrt{1600}$$

- 4×10

Answer: 40

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.09

Prime and composite numbers

Date _____

Learning intentions

We are learning to:

- identify prime and composite numbers

Success criteria

I can:

- ☐ know a prime has exactly 2 factors
- ☐ know a composite number has more than 2 factors
- ☐ list the primes and composites in a range

Key terms Write the definition as your teacher explains it.**Factor****Prime****Composite**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

List the factors of 12 and of 13.

Example 2

Is 1 prime, composite or neither?

YOU DO

On your own: Set A. List all the factors of each number.

1 6

4 20

2 11

5 17

3 15

6 36

YOU DO On your own: Set B. Prime or composite?

1 7

4 39

2 21

5 47

3 29

6 91

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** List the prime numbers less than 30.**Example 4** Is 51 prime? Explain.**YOU DO** On your own: Set C. List the numbers.

1 The prime numbers between 50 and 70.

3 The composite numbers larger than 40 but less than 59.

2 The composite numbers between 20 and 32.

4 The composite numbers between 45 and 70 that are odd.



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** List the prime numbers between 26 and 50.
Example 6 List the composite numbers between 45 and 60.
YOU DO On your own: Set D. Think it through.**1** Why is 2 the only even prime number?
3 Twin primes differ by 2, like 11 and 13. Find all twin primes under 50.
2 Find two primes that add to 36 (find 2 ways).
4 Is $2 \times 3 \times 5 + 1$ prime?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Is 23 prime or composite?

Level 2

List the factors of 28.

Level 3

List the primes between 30 and 45.

Extension · Stretch yourself

Worked example

Is 143 prime?

1. Only test primes up to $\sqrt{143} \approx 12$: 2, 3, 5, 7, 11
2. $143 \div 11 = 13$

Answer: No: $143 = 11 \times 13$

YOU DO

On your own: read the worked example, then try these.

E1 Is 197 prime? Show your tests.

E2 Is 221 prime?

Lesson summary

Missed the lesson? Start here.

Remember

1. A **prime** has exactly 2 factors: 1 and itself.
2. A **composite** number has more than 2 factors.
3. 1 is neither prime nor composite.
4. Test for primes by dividing by 2, 3, 5, 7, ...

Level 1

Factors of 12

1. 1×12 , 2×6 , 3×4

Answer: 1, 2, 3, 4, 6, 12

Level 2

Is 51 prime?

1. $5 + 1 = 6$ so divisible by 3

Answer: no, $51 = 3 \times 17$

Level 3

Primes 26 to 50

1. Test each odd number

Answer: 29, 31, 37, 41, 43, 47

Extension

Is 143 prime?

1. Try 11

Answer: no, 11×13

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.10

Prime factors

Date _____

Learning intentions

We are learning to:

- write a number as a product of prime factors using a factor tree

Success criteria

I can:

- ☐ draw a factor tree until every branch ends in a prime
- ☐ write the answer in index notation
- ☐ use prime factors to find square and cube roots

Key terms Write the definition as your teacher explains it.**Factor tree****Prime factor**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Use a factor tree for 60.

Example 2

Use a factor tree for 200.

YOU DO

On your own: Set A. Draw a factor tree for each number.

1 12

3 30

5 45

2 18

4 40

6 50

YOU DO

On your own: Set B. Write each number as a product of primes in index notation.

1 24

2 36

3 72

**Level 2 · Building up****WE DO**

With your teacher: copy what they write on the board.

Example 3

Use a factor tree for 144.

Example 4

Write 90 as a product of primes in index notation.

YOU DO

On your own: Set C. Use a factor tree. Write in index notation.

1 60

3 144

5 675

2 200

4 252

6 392



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Use a factor tree to find $\sqrt{256}$.
Example 6 Use a factor tree to find $\sqrt[3]{5832}$.
YOU DO On your own: Set D. Use prime factors to evaluate each root.**1** $\sqrt{256}$ **3** $\sqrt[3]{5832}$ **2** $\sqrt{324}$ **4** $\sqrt[3]{2744}$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write 20 as a product of primes.

Level 2

Write 84 as a product of primes in index notation.

Level 3Use prime factors to find $\sqrt{576}$.

Extension · Stretch yourself

Worked example

What is the smallest number that 72 must be multiplied by to make a perfect square?

- $72 = 2^3 \times 3^2$
- A square needs even indices
- $\times 2$ gives $2^4 \times 3^2 = 144$

Answer: 2

YOU DO

On your own: read the worked example, then try these.

- E1** Find the smallest number that 180 must be multiplied by to make a perfect square.

- E2** Find the smallest number that 18 must be multiplied by to make a perfect cube.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Split the number into two factors.
 2. Keep splitting until every branch is prime.
 3. Write the primes in order, using indices.
 4. For $\sqrt{}$, halve the indices; for $\sqrt[3]{}$, divide them by 3.

Level 1

Factor tree for 60

- $60 = 6 \times 10 = 2 \times 3 \times 2 \times 5$

Answer: $2^2 \times 3 \times 5$

Level 2

200

- $200 = 2 \times 100 = 2 \times 10 \times 10$

Answer: $2^3 \times 5^2$

Level 3

$\sqrt{256}$

- $256 = 2^8$
- Halve the index: 2^4

Answer: 16

Extension

$72 \times ? = \text{square}$

- $2^3 \times 3^2$

Answer: 2

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LESSON
3.11

Highest common factor

Date _____

Learning intentions

We are learning to:

- find the highest common factor (HCF) of two numbers

Success criteria

I can:

- ☐ list the factors of each number and find the largest common one
- ☐ use prime factors to find the HCF
- ☐ use the HCF to solve problems

Key terms Write the definition as your teacher explains it.**Common factor****HCF**

Notes

Level 1 · Getting started**WE DO** With your teacher: copy what they write on the board.**Example 1** Find the HCF of 12 and 18.**Example 2** Find the HCF of 20 and 48.**YOU DO** On your own: Set A. Find the HCF by listing factors.

1 6 and 9

2 8 and 12

3 10 and 15

4 14 and 21

5 16 and 24

6 9 and 20

YOU DO On your own: Set B. Find the HCF.

1 20 and 48

4 24 and 40

2 36 and 84

5 18 and 54

3 45 and 60

6 28 and 42

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Find the HCF of 36 and 84.

Example 4 Find the HCF of 17 and 34.

YOU DO On your own: Set C. Use factor trees to find the HCF.

1 243 and 162

3 96 and 144

5 150 and 225

2 120 and 200

4 84 and 126

6 98 and 140



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Use factor trees to find the HCF of 243 and 162.
Example 6 Use factor trees to find the HCF of 180 and 252.
YOU DO On your own: Set D. HCF problems.**1** Simplify $36/84$ using the HCF.
3 Two ribbons are 45 cm and 75 cm long. They are cut into equal pieces as long as possible. How long is each piece?
2 24 boys and 32 girls are put into equal teams with no mixing. What is the largest team size?
4 The HCF of two numbers is 6. Both are between 20 and 40. Find a possible pair.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the HCF of 8 and 20.

Level 2

Find the HCF of 42 and 70.

Level 3

Use prime factors to find the HCF of 168 and 252.

Extension · Stretch yourself

Worked example

Find the HCF of 36, 60 and 84.

- $36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$, $84 = 2^2 \times 3 \times 7$
- Common primes: $2^2 \times 3$

Answer: 12

YOU DO

On your own: read the worked example, then try these.

- E1** Find the HCF of 48, 72 and 120.

- E2** Find the HCF of 105, 165 and 195.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. List the factors of both numbers.
 3. The largest one is the **HCF**.

2. Circle the common factors.
4. Or: multiply the prime factors they share.

Level 1

HCF of 12 and 18

1. 12: 1, 2, 3, 4, 6, 12
2. 18: 1, 2, 3, 6, 9, 18

Answer: 6

Level 2

HCF of 20 and 48

1. Common: 1, 2, 4

Answer: 4

Level 3

HCF of 243 and 162

1. $243 = 3^5$
2. $162 = 2 \times 3^4$

Answer: $3^4 = 81$

Extension

HCF of 36, 60, 84

1. $2^2 \times 3$

Answer: 12

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.12

Lowest common multiple

Date _____

Learning intentions

We are learning to:

- find the lowest common multiple (LCM) of two numbers

Success criteria

I can:

- ☐ list multiples of each number and find the smallest common one
- ☐ use prime factors to find the LCM
- ☐ use the LCM to solve problems

Key terms Write the definition as your teacher explains it.**Multiple****LCM**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find the LCM of 4 and 6.**Example 2** Find the LCM of 6 and 8.**YOU DO** On your own: Set A. List the first 6 multiples of each number.

1 3

4 12

2 5

5 9

3 8

6 15

YOU DO On your own: Set B. Find the LCM.

1 2 and 3

4 3 and 7

2 4 and 10

5 8 and 12

3 6 and 9

6 10 and 15

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Find the LCM of 18 and 10.**Example 4** Find the LCM of 5 and 7.**YOU DO** On your own: Set C. Find the LCM. Use factor trees for larger numbers.

1 18 and 10

3 12 and 90

5 24 and 36

2 25 and 16

4 14 and 21

6 20 and 45



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Use factor trees to find the LCM of 25 and 16.
Example 6 Use factor trees to find the LCM of 12 and 90.
YOU DO On your own: Set D. LCM problems.

- 1 Buses leave at 6:00 am. One route runs every 12 minutes and another every 15 minutes. When do they next leave together?

- 3 Hot dogs come in packs of 10 and buns in packs of 8. What is the smallest number of each you can buy to have no leftovers?

- 2 Find the lowest common denominator of $\frac{5}{6}$ and $\frac{3}{8}$.

- 4 The LCM of two numbers is 24 and one number is 8. Find all possible values of the other number.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the LCM of 3 and 4.

Level 2

Find the LCM of 12 and 16.

Level 3

Use prime factors to find the LCM of 28 and 42.

Extension · Stretch yourself

Worked example

Find the LCM of 4, 6 and 10.

- $4 = 2^2$, $6 = 2 \times 3$, $10 = 2 \times 5$
- Highest power of each prime: $2^2 \times 3 \times 5$

Answer: 60

YOU DO

On your own: read the worked example, then try these.

- E1** Find the LCM of 6, 8 and 9.

- E2** Three lights flash every 4, 5 and 6 seconds. They flash together now. When do they next flash together?

Lesson summary

Missed the lesson? Start here.

- Remember**
- List multiples of each number.
 - The smallest common one is the **LCM**.
 - Prime factors: take the highest power of each prime.
 - $\text{LCM} \times \text{HCF} = \text{the product of the two numbers.}$

Level 1

LCM of 4 and 6

- 4, 8, 12
- 6, 12

Answer: 12

Level 2

LCM of 18 and 10

- 18, 36, 54, 72, 90
- 10, ..., 90

Answer: 90

Level 3

LCM of 25 and 16

- 5^2 and 2^4
- No common primes: multiply

Answer: 400

Extension

LCM of 4, 6, 10

- $2^2 \times 3 \times 5$

Answer: 60

This space is left blank on purpose. It is the back of your exit ticket.